# L2.6b More Optimization

§3.7 — build the objective, then choose the test

## The First Derivative Test for Absolute Extreme Values

Neither of these hands you the thing to optimize — write it down first, then run the template. Then name the approach that finished the problem, and the feature of the interval that chose it.

1. **Two parabolas.** What is the minimum vertical distance between the parabolas $y = x^2 + 1$ and $y = x - x^2$?

2. **The far side of an ellipse.** Find the points on the ellipse $4x^2 + y^2 = 4$ that are farthest away from the point $(1, 0)$.

   **Stop.** Before you differentiate anything: what values can $x$ take on this ellipse, and where did that come from? And is the question asking for the point, or for the distance?

## The optimization template on an open interval

**Stop.** Every problem from here on carries a letter that is *not* the variable — $a$, $E$, $r$, a given area, a given perimeter. They are **constants.** Name which letter is varying before you differentiate: what is $\dfrac{d}{dR}\, r^2$?

3. **An isosceles triangle.** The two equal sides of an isosceles triangle have length $a$. Find the length of the third side that maximizes the area of the triangle.

**4. Power in a resistor.** A resistor of $R$ ohms is connected across a battery of $E$ volts with internal resistance $r$ ohms. The power, in watts, in the external resistor is

$$P = \frac{E^2 R}{(R + r)^2}.$$

If $E$ and $r$ are fixed but $R$ varies, what is the maximum value of the power?

> **Stop.** There is no picture to draw here. What replaces it — what has to be written down before you can differentiate?

## At the boards

**5.** BOARDS **The square theorem.** Half the room takes (a), half takes (b).

**(a)** Show that of all the rectangles with a given **area**, the one with smallest perimeter is a square.

**(b)** Show that of all the rectangles with a given **perimeter**, the one with greatest area is a square.

There is no number anywhere in either one. Name the given quantity, carry it through, and finish with the two side lengths in terms of it. When both walls are up, put the intervals side by side — they are not the same kind, and the justification differs because of it.

**6.** BOARDS **A cylinder in a sphere.** A right circular cylinder is inscribed in a sphere of radius $r$. Find the largest possible **surface area** of such a cylinder.

Name every letter in your picture before you write a slot, and say which one is fixed.